Electronic Signatures Explained

Simple, Advanced, and Qualified Signatures Compared

PDF QES Signer | A brief introduction to the eIDAS Regulation (EU) No. 910/2014

Contents

1 Introduction

The EU **eIDAS** Regulation (Regulation (EU) No. 910/2014) distinguishes three levels of electronic signatures. They differ not in the underlying *technology*, but above all in their **legal effect** and their **evidentiary weight** in court: Who was identified, who issues the certificate, and how reliably is it excluded that someone else signed?

All three levels are legally valid throughout the EU – but only the highest level, the **Qualified Electronic Signature (QES)**, is legally equivalent to a handwritten signature.

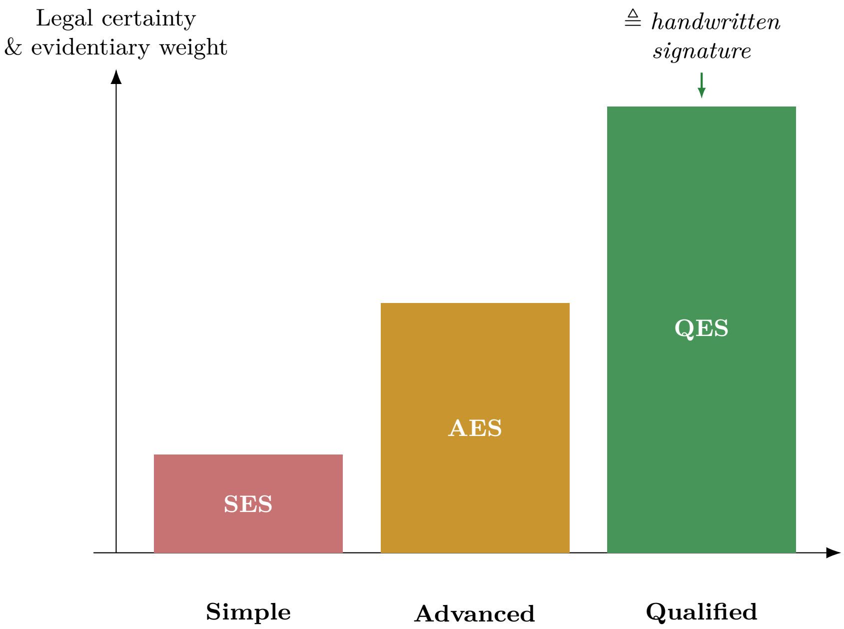

Figure 1. *The higher the level, the stronger the legal evidentiary weight. Only the Qualified Electronic Signature (QES) is legally equivalent to a handwritten signature.*

2 Simple Electronic Signature (SES)

Definition (Art. 3(10) eIDAS): Data in electronic form which is attached to or logically associated with other data in electronic form and which is used by the signatory to sign.

Typical examples:

- A scanned or inserted image of a signature in a PDF
- A typed name at the bottom of an email
- Clicking “I agree” / “Place order”

Legal effect: A simple electronic signature may not be denied legal effect and admissibility as evidence in court solely on the grounds that it is in electronic form (Art. 25(1) eIDAS). However, there is **no statutory presumption of authenticity** – in a dispute, whoever relies on the signature must actively prove that it genuinely originates from the named person. In practice this is difficult, since the signature can be freely copied and was created without any identity verification.

Technology: As a rule, there is **no** cryptographic protection here. An image of a signature or a typed name is simply additional content of the file – nothing prevents the text from being changed afterwards or the “signature” from being copied into a different document. Neither the identity of the signatory nor the integrity of the document is technically guaranteed.

Security level: low – no identity verification, no tamper detection.

3 Advanced Electronic Signature (AES)

Definition (Art. 26 eIDAS): An advanced electronic signature must simultaneously meet four requirements:

1. It is **uniquely linked to the signatory**.
2. It is **capable of identifying the signatory**.
3. It was created using electronic signature creation data that the signatory can, with a **high level of confidence, use under their sole control**.
4. It is **linked to the signed data** in such a way that any subsequent change to the data is **detectable**.

Typical examples: Signature services such as DocuSign or Adobe Sign with email and SMS verification, a signature captured with signature dynamics (pressure, speed) on a signature pad, or a signed PDF using a personal (non-qualified) certificate.

Legal effect: Stronger evidentiary weight than a simple signature, since identity and integrity are technically verifiable. However, it still does **not** replace statutory written-form requirements (e.g. § 126 BGB in Germany) and is not automatically equivalent to a handwritten signature.

Technology: The advanced signature uses asymmetric cryptography (public-key methods): a hash value is computed over the document content and encrypted with the signatory's **private key** (digital signature, e.g. RSA or ECDSA). Any subsequent change to the document changes the hash value – the signature then becomes invalid, and the tampering is detectable. Identity derives from the certificate associated with the key; how reliable that association is depends on how rigorously the issuer (e.g. a company or email provider) verified the identity before issuing it.

Multiple signatures: If several people need to sign the same PDF, no existing signature is broken to do so. Each additional signature creates a new **revision** of the document (an incremental update) and additionally covers all previously existing signatures as well as the content. Earlier signatures remain untouched and stay valid – new signatures are added purely **additively**, revision by revision.

Security level: medium – tamper detection is present, but identity is typically not verified by an independent, trusted authority.

4 Qualified Electronic Signature (QES)

Definition (Art. 3(12) eIDAS): An advanced electronic signature that is additionally

- created using a **qualified certificate**, issued by a **qualified trust service provider (QTSP)** who has verified the signatory's identity beforehand beyond doubt (e.g. via video identification or an ID check), and
- created using a **qualified electronic signature creation device (QSCD)** – specially secured hardware such as a signature card or a certified remote-signing HSM.

Legal effect (Art. 25(2) eIDAS): The QES is **legally equivalent to a handwritten signature in all EU member states**. The **statutory presumption of authenticity** applies: in a dispute, the signatory does not have to prove authenticity – rather, the other party would have to prove forgery (reversal of the burden of proof). Only the QES can replace the statutory written-form requirement.

Typical examples: Qualified remote signature services such as D-Trust sign-me, Swisscom All-in Signing Service, A-Trust, or the German Bundesdruckerei – **PDF QES Signer** uses exactly this kind of service to sign documents.

Technology: Cryptographically, the QES works like the AES (hash value + digital signature with a private key), additionally secured by three factors: the private key never leaves the certified hardware (QSCD) and cannot be exported. The associated certificate is issued only after the QTSP has verified the identity beyond doubt. For PDF documents (PAdES standard), the signature is computed over a precisely defined byte range of the file and supplemented with a qualified timestamp – so the signature remains verifiable even years later, even if the signing certificate has since expired.

Security level: high – verified identity, secured hardware, tamper detection, statutory presumption of validity.

5 Comparison at a Glance

Feature	Simple (SES)	Advanced (AES)	Qualified (QES)
Identity verification	×	(internal to provider)	✓ by QTSP
Tamper detection	×	✓	✓
Qualified certificate	×	×	✓
Secure hardware (QSCD)	×	×	✓
Equivalent to handwritten signature	×	×	✓
Evidentiary weight in court	low	medium	high (presumption)
Replaces statutory written form (§126 BGB)	×	×	✓

Table 1. *Comparison of the three eIDAS signature levels.*

6 Which Signature Do I Need?

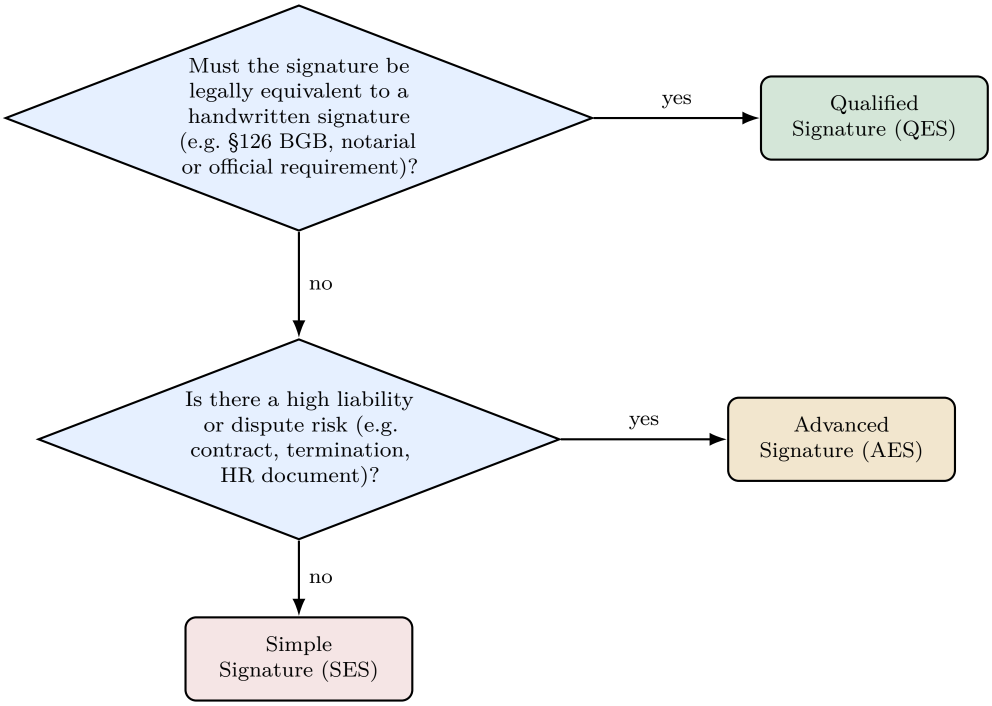

Figure 2. *Simplified decision guide. When in doubt, or where there is legal uncertainty, the higher level is recommended.*

7 Relation to PDF QES Signer

PDF QES Signer creates qualified electronic signatures (PAdES, including long-term archiving per PAdES-B-LTA) via qualified remote signature services. Every signature it creates automatically satisfies all requirements of the highest eIDAS level: identity verified by the trust service provider, a qualified certificate, and creation on certified hardware (QSCD) at the provider.

Source code, downloads, and further documentation: codeberg.org/pitbo/pdf-qes-signer.

8 Glossary

eIDAS – *electronic IDentification, Authentication and trust Services*; EU Regulation (EU) No. 910/2014 governing electronic identification and trust services in the internal market.

QTSP – *Qualified Trust Service Provider*; a trust service provider accredited and supervised by a national supervisory body, authorized to issue qualified certificates.

QSCD – *Qualified electronic Signature Creation Device*; certified, specially secured hardware (a smart card or a remote-signing HSM) used to create qualified signatures.

PAdES – *PDF Advanced Electronic Signatures*; the ETSI standard for electronic signatures in PDF documents, on which PDF QES Signer is built.